ECA1803 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO5 – Case Study Based Assignment

DWDM Metro Link Commissioning: From Field Measurements to a Standards-Based Upgrade Decision

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Stage 1 – Interpretation of the Measurement / Performance Requirement

1.1 Instruments and what each one validates

Instrument	Quantity validated on this link
Optical power meter with 1550 nm source	End-to-end insertion loss, received power per channel, and hence the power margin against −24 dBm.
Optical spectrum analyser (OSA)	Channel frequency on the ITU grid, per-channel power, EDFA ASE level and OSNR (0.1 nm reference bandwidth).
OTDR (1550 nm)	Fibre length and attenuation, and the position, loss and reflectance of every splice and connector.
Eye / BER tester	Eye opening, extinction ratio, jitter (DJ, RJ), Q-factor and BER of each channel.
Dispersion / PMD analyser	Chromatic dispersion (D) and PMD coefficient, which confirm the dispersion penalty and the PMD tolerance.

1.2 Components of the practical budgets

Power budget: fibre attenuation (60 km × 0.21 dB/km), connector loss, splice loss and the demultiplexer loss (6 dB). **Penalties:** chromatic dispersion penalty, extinction-ratio (ER) penalty, PMD allowance and the effect of reflections from poor connectors. **Noise:** EDFA ASE (set by noise figure and input power, giving OSNR) and receiver noise. On top of these a design margin of at least 3 dB has to remain for ageing and repairs.

1.3 Eye opening, jitter, extinction ratio, Q and BER

$Q = (\mu_1 - \mu_0)/(\sigma_1 + \sigma_0)$ and $BER = \tfrac{1}{2}\,\mathrm{erfc}(Q/\sqrt{2})$. A low extinction ratio raises the "0" level μ_0 and shrinks $\mu_1 - \mu_0$, so the eye closes vertically and Q falls. Jitter (DJ + RJ) closes the eye horizontally, so the sampling instant has less room. Both effects reduce the margin at the decision point. For $BER = 10^{-12}$ we need $Q \geq 7.03$, and the receiver sensitivity of −24 dBm is only valid when $ER \geq 8.2$ dB.

1.4 OTDR limitations

The dynamic range (DR) decides whether the far end of the link is visible, since it must cover the one-way trace loss plus 6 dB. The event dead zone (EDZ) is the shortest distance after a reflective event at which a second reflective event can still be seen separately, and the attenuation dead zone (ADZ) is the distance needed before the loss of the next event can be measured. A longer pulse gives more DR but larger dead zones, so the pulse width is always a trade-off. Here, events 3 and 4 are only 9 m apart, which is the critical case.

1.5 Standards, splices and connectors

ITU-T G.652.D is the standard single-mode fibre with low water peak and low PMD, which suits both 10G NRZ and coherent transmission at 1550 nm. ITU-T G.694.1 defines the DWDM frequency grid (12.5, 25, 50 and 100 GHz fixed spacing, and the flexible grid with slot width = 12.5 × m GHz) anchored at 193.1 THz. Fusion splices give about 0.05 dB and almost no reflection, whereas mechanical splices give higher loss and measurable reflectance. For connectors, SC (2.5 mm ferrule) is large, LC and CS use a 1.25 mm ferrule and are compact duplex types, and MPO is a multi-fibre connector used for parallel optics. UPC or APC polish decides the reflectance.

1.6 Practical constraints

- The same fibre pair must be used, so there is no re-laying of fibre.
- System margin must be at least 3 dB, and the loss must be brought back to the 21.25 dB design value.
- The upgrade to 1.6 Tb/s must fit in the C-band on the G.694.1 grid and must use as little power per bit as possible.

Performance Requirement Statement: After repair and upgrade, the 60 km G.652.D link must have a total loss of not more than 21.25 dB, no OTDR event above its allowance (connector 0.5 dB, fusion splice 0.05 dB), a system margin of at least 3 dB on every channel after penalties, ER ≥ 8.2 dB and Q ≥ 7.03 (BER ≤ 10^{-12}), and it must carry 1.6 Tb/s on the same fibre pair with a G.694.1 channel plan, OSNR above the requirement of the chosen format and the lowest energy per bit.

Stage 2 – Link Budget, Dispersion and OSNR Configuration

2.1 Loss budget: design and as-built

Fibre attenuation = 60 × 0.21 = 12.6 dB. For the as-built case the power-meter and OTDR values replace the allowances. There are four connectors: three measured at 0.5 dB (including event 4) and event 3 at 1.20 dB. There are thirteen splices: twelve fusion splices at 0.05 dB and event 2 (mechanical) at 0.62 dB.

Table 2.1 – Design and as-built link loss

Component	Design (dB)	As-built (dB)
Fibre (60 km × 0.21 dB/km)	12.60	12.60
Connectors	4 × 0.5 = 2.00	3 × 0.5 + 1.20 = 2.70
Splices	13 × 0.05 = 0.65	12 × 0.05 + 0.62 = 1.22
Demultiplexer	6.00	6.00
Total link loss	**21.25**	**22.52**

The as-built link has 22.52 − 21.25 = **1.27 dB excess loss**. Of this, event 3 contributes 1.20 − 0.50 = 0.70 dB and event 2 contributes 0.62 − 0.05 = 0.57 dB. Event 3 is therefore the dominant excess loss.

2.2 Received power, penalties and margin

Received power = launch power − loss = +1 − 22.52 = −21.52 dBm. Dispersion penalty: $B \cdot D \cdot L \cdot \sigma\lambda = 10\times10^{9} \times 17\times10^{-12} \times 60 \times 0.03 = 0.306$, so $\delta = 5 \log_{10}[1 + 2\pi(0.306)^2] = 5 \log_{10}(1.588)$ = **1.00 dB**. Extinction-ratio penalty: $\delta ER = 10 \log_{10}[(r + 1)/(r - 1)]$, where r is the linear ER. For Channel 3, r = 10^0.9 = 7.94 giving **1.10 dB**. For Channel 5, r = 10^0.45 = 2.82 giving **3.22 dB**.

Table 2.2 – Received power and system margin (as-built)

Quantity	Channel 3	Channel 5
Launch power (dBm)	+1.00	+1.00
As-built loss (dB)	22.52	22.52
Received power (dBm)	−21.52	−21.52
Power-only margin vs −24 dBm (dB)	2.48	2.48
Dispersion penalty (dB)	1.00	1.00
ER penalty (dB)	1.10	3.22
Effective received power (dBm)	−23.62	−25.74
System margin vs −24 dBm (dB)	**0.38**	**−1.74**
Required margin (dB)	3.00	3.00
Result	Fails (short by 2.62 dB)	Fails (short by 4.74 dB)

Even before the penalties are counted, the power-only margin is 2.48 dB, which is already below 3 dB. This matches the power-meter observation in the case. With the penalties, Channel 3 is left with only 0.38 dB and Channel 5 is below sensitivity. Also, Channel 5 has ER = 4.5 dB, which is below 8.2 dB, so the −24 dBm sensitivity does not even apply to it and the real situation is worse than the table shows.

2.3 OSNR and PMD

OSNR = 58 + P_in − NF = 58 + (−6) − 5 = **47 dB** (0.1 nm). The requirement is 20 dB, so the noise margin is 27 dB. Because the booster is at the transmit end and the loss that follows reduces signal and ASE equally, this OSNR is preserved at the receiver. The link is therefore limited by **power and penalties, not by noise**.

PMD = 0.1 × √60 = 0.775 ps. The tolerance is 10% of the 100 ps bit period = 10 ps, so PMD uses only 0.8% of the bit period and is not a concern.

2.4 Corrective action

Repairing the two excess-loss events brings the loss back to 21.25 dB, but with the 2.10 dB penalty of a healthy channel (1.00 + 1.10) the margin is still only 1.65 dB. A launch-power increase is also needed.

Table 2.3 – Effect of corrective actions (healthy channel, penalties 2.10 dB)

Case	Loss (dB)	Rx power (dBm)	Effective Rx (dBm)	Margin (dB)
As-built	22.52	−21.52	−23.62	0.38
Event 3 repaired (1.20 → 0.50 dB)	21.82	−20.82	−22.92	1.08
Event 2 repaired (0.62 → 0.05 dB)	21.95	−20.95	−23.05	0.95
Events 2 and 3 repaired	21.25	−20.25	−22.35	1.65
Repairs + launch +3 dBm per channel	21.25	−18.25	−20.35	**3.65**

At least 1.35 dB more launch power is needed. I take +3 dBm per channel (+2 dB), which gives 3.65 dB margin with 0.65 dB spare for ageing. The total booster output for eight channels is about +12 dBm, which is acceptable, and the OSNR is not affected. The Channel 5 transmitter must also be corrected (Stage 4), otherwise its extra 2.12 dB of ER penalty remains.

Design Decision 1: I recommend repairing the link first (event 3 first, then event 2) and then raising the launch power to +3 dBm per channel, because the as-built link has 22.52 dB loss (1.27 dB over design) and only 0.38 dB margin on Channel 3 against the 3 dB requirement, while OSNR is 47 dB (27 dB above requirement) and PMD is only 0.775 ps. The limit is excess loss and penalties, not noise, and the repairs plus +2 dB of launch power give a margin of 3.65 dB.

Stage 3 – OTDR, Splice and Connector Configuration

3.1 Pulse width and test arrangement

One-way trace loss (fibre only, since the demultiplexer is not in the trace) = 12.6 + 2.70 + 1.22 = 16.52 dB. Required dynamic range = 16.52 + 6 = **22.52 dB**. Spatial pulse length = $c \cdot \tau/(2n)$, with $c/n = 2.998\times10^{8}/1.4682 = 2.042\times10^{8}$ m/s. The two reflections of events 3 and 4 are separated in time by $2n\Delta L/c = 2 \times 1.4682 \times 9/2.998\times10^{8} = 88$ ns.

Table 3.1 – OTDR pulse width selection

Pulse	DR (dB)	DR ≥ 22.52 dB?	Spatial length	EDZ / ADZ	Events 3 and 4 separate?	Loss of event 4 measurable?
10 ns	20	No	1.02 m	1 m / 4 m	Yes	Yes (ADZ 4 m < 9 m)
100 ns	27	Yes (4.5 dB spare)	10.2 m	5 m / 15 m	Yes (EDZ 5 m < 9 m)	No (ADZ 15 m > 9 m)
1 µs	33	Yes	102 m	40 m / 100 m	No, merged	No

The 10 ns pulse cannot see the whole link, and the 1 µs pulse merges events 3 and 4. So the **100 ns pulse** is used for the full-length trace (it locates events 1, 2 and 3 and resolves the event 4 peak). For the loss of the second connector, a **10 ns trace is taken from the far end**, where events 3 and 4 lie only about 91 m and 100 m from the instrument, so the 20 dB dynamic range is more than enough. The test arrangement is a launch cable and a receive cable (about 500 m to 1 km each) so that the first and last connectors fall outside the instrument's own dead zone, with traces taken in both directions and averaged for the true splice loss.

3.2 Event interpretation

Table 3.2 – OTDR events

Event	Location	Loss / reflectance	Interpretation
1	12.00 km	0.05 dB, non-reflective	Good fusion splice. No action.
2	31.50 km	0.62 dB, −34 dB	Mechanical splice. The loss is about 12 times the fusion allowance and the reflection comes from the gap and index-matching gel. Re-splice by fusion.
3	59.90 km	1.20 dB, −22 dB	LC/UPC patch-panel connector in poor condition. A UPC connector should be better than about −50 dB, so −22 dB points to a dirty or scratched end face, an air gap or misalignment. Clean and inspect, or replace.
4	59.909 km	−40 dB, loss not resolved	Second connector of the 9 m patch cord. Its loss (0.5 dB by power meter) is within allowance. The pair of connectors with 9 m of fibre between them also forms a reflecting cavity.

Table 3.3 – Fusion splice compared with mechanical splice

Parameter	Fusion splice	Mechanical splice
Typical loss	0.02 to 0.1 dB (0.05 dB here)	0.2 to 1 dB (0.62 dB here)
Reflectance	Practically none (non-reflective)	Reflective (−34 dB here)
Long-term stability	Permanent, very stable	Gel and alignment can change with temperature
Suitability for upgrade	Best for high-rate and coherent links	Acceptable only for temporary or emergency repair

3.3 Round-trip delay and location uncertainty

Round-trip delay $t = 2nL/c$, and the location error from a group-index error is $\Delta L = L \times \Delta n/n = L \times 0.002/1.4682 = 0.136\%$ of L.

Table 3.4 – Round-trip delay and uncertainty

Event	Distance	Round-trip delay (µs)	Uncertainty for ±0.002 in n
1	12.000 km	117.53	± 16.3 m
2	31.500 km	308.53	± 42.9 m
3	59.900 km	586.69	± 81.6 m
4	59.909 km	586.78	± 81.6 m

The absolute position of event 2 is uncertain by about ±43 m, so it must be located using the cable sheath markings and the bidirectional traces before the closure is opened. Event 3 is at a known patch panel, so the error does not matter there. The 9 m spacing between events 3 and 4 is hardly affected (error about ±0.012 m) because the error scales with distance.

3.4 Connector selection for the patch-panel repair

The link uses a single duplex fibre pair at the patch panel. SC is bulky, MPO is meant for multi-fibre parallel optics and typically has higher insertion loss, and CS is only needed for very high-density pluggable ports. **LC** is compact, duplex and already used on this panel, so it is the practical choice. For the polish, APC (8° angled end face) gives reflectance of about −60 dB or better, compared with about −50 dB for UPC, while insertion loss is similar (about 0.2 to 0.3 dB when clean) and inside the 0.5 dB allowance. Since events 3 and 4 form a reflecting cavity near the receiver end, I select **LC/APC** for the repaired patch cords, provided the demultiplexer and equipment ports are also APC (APC must never be mated with UPC). If the equipment is UPC, new LC/UPC cords with a cleaned and inspected end face are used.

3.5 Repair priority and verification

Priority	Event	Reason	Verification
1	Events 3 and 4 (patch panel)	Largest excess loss (0.70 dB), worst reflectance (−22 dB), easy and quick to fix without cable access.	OTDR 10 ns from the far end: event 3 ≤ 0.5 dB, reflectance ≤ −50 dB (UPC) or −60 dB (APC); power meter shows the connector pair back at 0.5 dB each.
2	Event 2 (31.5 km)	Excess loss 0.57 dB and a reflective splice; needs cable access and a new fusion splice.	OTDR 100 ns, both directions averaged: splice ≤ 0.05 dB and non-reflective.
Final	Whole link	Confirm the budget.	Power meter end-to-end loss ≤ 21.25 dB, and full OTDR trace with no event above allowance.

Design Decision 2: I recommend giving the highest repair priority to events 3 and 4 (the patch-panel connectors), replacing them with LC/APC (or cleaned LC/UPC) and checking them with a 10 ns OTDR trace from the far end, then re-splicing event 2 by fusion and checking it with a 100 ns bidirectional trace, because event 3 has the largest excess loss (0.70 dB) and by far the worst reflectance (−22 dB), the 100 ns pulse (DR 27 dB against 22.52 dB needed, EDZ 5 m) is the only one that covers the full link and separates events 3 and 4, and the ADZ of 15 m means the second connector's loss can only be measured with the 10 ns pulse.

Stage 4 – Eye Diagram, BER and Quantum-Limit Analysis

4.1 BER of the two channels

Channel	Q	BER = ½ erfc(Q/√2)	Against 10^{-12} target
Channel 3	7.6	1.5×10^{-14}	Meets (Q ≥ 7.03)
Channel 5	4.7	1.3×10^{-6}	Fails by about six orders of magnitude

4.2 Total jitter and eye width

TJ = DJ + 14.07 × RJ. The eye width is 1 − TJ (in UI), and 1 UI = 100 ps at 10 Gb/s.

Channel	DJ (UI)	RJ (UI rms)	TJ at 10^{-12} (UI)	Eye width (UI)	Eye width (ps)
Channel 3	0.18	0.012	0.349	0.651	65.1
Channel 5	0.30	0.022	0.610	0.390	39.0

Q and ER describe the **vertical (noise/amplitude) margin**, and the eye width describes the **horizontal (timing) margin**. Channel 5 is poor on both: its Q is low and its eye width is only 0.39 UI against 0.65 UI for Channel 3, because both its DJ and RJ are larger. Channel 3 has comfortable margin in both directions.

4.3 Is the Channel 5 problem caused by its extinction ratio?

The Q-factor scales with (r − 1)/(r + 1): 0.776 for Channel 3 and 0.476 for Channel 5, so the expected ratio Q_5/Q_3 is 0.476/0.776 = 0.613. The measured ratio is 4.7/7.6 = 0.618, which agrees within 1%. In dB, the measured Q ratio is $10 \log_{10}(7.6/4.7) = 2.09$ dB, and the ER penalty difference is 3.22 − 1.10 = 2.12 dB. The low ER of Channel 5 therefore explains almost the entire Q difference.

Both channels travel in the same fibre, so the loss, the dispersion penalty (1.00 dB) and the fibre reflections are common to them. Channel 3 works with these, so the **root cause for Channel 5 is channel-specific: its transmitter** (ER 4.5 dB, higher DJ and RJ, probably a wrong bias or modulator setting or an ageing laser). The fibre faults of Stage 3 are a **shared cause** that reduces the margin of both channels, but they do not explain the difference between them.

4.4 Quantum limit

For ideal direct detection, BER = ½ exp(−N_1) = 10^{-12}, so $N_1 = \ln(5 \times 10^{11})$ = **26.9 photons per 1-bit**. At 1550 nm, ν = c/λ = 1.934×10^{14} Hz and hν = 1.282×10^{-19} J. The power is P = $N_1 \cdot h\nu \cdot B$ = $26.94 \times 1.282 \times 10^{-19} \times 10^{10} = 3.45 \times 10^{-8}$ W = 34.5 nW = **−44.6 dBm**. The receiver sensitivity of −24 dBm is therefore **20.6 dB** away from the quantum limit. (If only half the bits are 1s, the average power at the limit is 3 dB lower, −47.6 dBm, and the gap is 23.6 dB.)

For coherent PSK at BER = 10^{-12}, the argument of erfc must be 4.974 (Q = 7.03). Homodyne: √(2N) = 4.974, so N = 12.4 photons per bit. Heterodyne: √N = 4.974, so N = 24.7 photons per bit.

Table 4.1 – Quantum limits at 10 Gb/s, BER 10^{-12}

Detection	Photons per bit	Sensitivity at 10 Gb/s	Gap from −24 dBm
Direct detection (ideal)	26.9	−44.6 dBm	20.6 dB
Homodyne PSK	12.4	−48.0 dBm	24.0 dB
Heterodyne PSK	24.7	−45.0 dBm	21.0 dB

Homodyne PSK needs 3.4 dB less power than direct detection and heterodyne is only 0.4 dB better, because heterodyne loses 3 dB to the image band. A real receiver is far from these limits, but they show that there is a lot of room to improve.

Implication for the coherent upgrade: with a strong local oscillator, a coherent receiver is shot-noise limited, so performance depends on OSNR and not on photon counting. The quantum limit grows in proportion to the bit rate. Using the homodyne figure of 12.4 photons per bit, the limit is −38.0 dBm at 100 Gb/s, −35.0 dBm at 200 Gb/s and −32.0 dBm at 400 Gb/s. The repaired link delivers about −20.3 dBm per channel at +1 dBm launch, which is about 11.7 dB above the 400G limit. Higher-order formats such as 16QAM need more photons per bit than this PSK limit, so the real margin is smaller, but it remains positive for all three options.

Design Decision 3: I recommend that Channel 5 is not accepted as it stands (Q = 4.7, BER 1.3 × 10^{-6}, ER 4.5 dB, eye width 0.39 UI) and that its transmitter is re-biased or replaced so that ER is at least 9 dB, because the Q ratio (0.618) matches the ER-factor ratio (0.613) and the 2.12 dB penalty difference matches the 2.09 dB Q difference, which proves a channel-specific transmitter fault while Channel 3 on the same fibre has Q = 7.6 (BER 1.5 × 10^{-14}). Recovery is confirmed when ER ≥ 8.2 dB, Q ≥ 7.03, TJ ≤ about 0.35 UI (eye width ≥ 0.65 UI like Channel 3) and BER ≤ 10^{-12}. The receiver is 20.6 dB from the quantum limit, so there is no fundamental obstacle.

Stage 5 – Integrated WDM Upgrade and Energy Strategy

5.1 ITU-T G.694.1 grid

Frequency of channel n on the 50 GHz grid = 193.1 THz + n × 0.05 THz. For n = +3, f = 193.1 + 0.15 = **193.25 THz**, and λ = c/f = 2.998×10^{8} / 193.25×10^{12} = **1551.36 nm**.

For Option Q, the Nyquist channel spacing is 32 GBd × (1 + 0.1) = 35.2 GHz. On the flex-grid with 12.5 GHz slots this needs 35.2/12.5 = 2.82, which rounds up to **3 slots = 37.5 GHz** per channel, so 8 channels need 24 slots (300 GHz).

5.2 Spectrum, spectral efficiency and OSNR margin

The repaired-link OSNR is 47 dB (Section 2.3). Occupied spectrum = number of channels × slot width. Payload spectral efficiency = 1.6 Tb/s / occupied spectrum.

Table 5.1 – Comparison of upgrade options for 1.6 Tb/s

Parameter	Option P	Option Q	Option R
Format	16 × 100G DP-QPSK	8 × 200G DP-16QAM	4 × 400G Si-photonic
Slot per channel	50 GHz (fixed)	37.5 GHz (3 slots)	75 GHz (fixed)
Occupied spectrum	800 GHz	300 GHz	300 GHz
Spectral efficiency	2.0 b/s/Hz	5.33 b/s/Hz	5.33 b/s/Hz
Required OSNR	12.5 dB	19.5 dB	26 dB
OSNR margin (47 dB available)	34.5 dB	27.5 dB	21.0 dB
Number of modules	16	8	4
Module power	22 W	20 W	15 W
Total power for 1.6 Tb/s	352 W	160 W	60 W
Energy per bit	220 pJ/bit	100 pJ/bit	**37.5 pJ/bit**

Energy per bit = module power / line rate, for example 15 W / 400 Gb/s = 37.5 pJ/bit. Option R uses 83% less energy per bit than P, 62.5% less than Q and 89% less than the legacy SFP+ (3.5 W / 10 Gb/s = 350 pJ/bit). Carrying 1.6 Tb/s with legacy SFP+ would need 160 modules (560 W) and 160 channels × 100 GHz = 16 THz (0.1 b/s/Hz), which does not fit in the C-band (about 4.4 THz), so it is not feasible anyway.

Role of Si-photonic integration: the modulators, waveguides and photodetectors are integrated on a CMOS-compatible silicon chip, together with the driver and DSP. This cuts coupling and packaging losses, drive power and size, and lets the module fit in a pluggable form. That is why a 400G module needs only 15 W, which is less than a 100G or 200G module. The price is a higher OSNR requirement (26 dB) and higher modulator insertion loss, but this link has 47 dB available, so this trade-off costs nothing here.

5.3 Sequence of work

The order is (1) baseline OTDR and power-meter test with end-face inspection, (2) repair events 3 and 4 and verify, (3) repair event 2 and verify, (4) correct the Channel 5 transmitter and check eye, ER and Q, (5) set the launch power, confirm loss ≤ 21.25 dB, margin ≥ 3 dB and OSNR on the OSA, (6) measure CD and PMD, and only then (7) install the upgrade. The repair comes first because the faults are cheap to fix and a reflective, lossy plant would give misleading results on the new coherent system. Also, the old 10G service should keep running at a healthy state until the new capacity is commissioned. The upgrade also needs flex-grid-capable filters or WSS in place of the fixed 100 GHz demultiplexer, but the fibre itself is not changed.

5.4 Ranking and selection

Criterion	Best	Second	Third
Spectrum	Q and R (300 GHz)	–	P (800 GHz)
OSNR margin	P (34.5 dB)	Q (27.5 dB)	R (21.0 dB)
Energy per bit	R (37.5 pJ/bit)	Q (100 pJ/bit)	P (220 pJ/bit)

All three options have a positive OSNR margin, so OSNR does not decide the choice, and the repaired link has 21 dB spare even for the most demanding Option R. Option R matches Q on spectrum and clearly beats it and P on energy, so **Option R is selected**. The 75 GHz slot corresponds to 6 flex-grid slots. A suitable channel plan on the 300 GHz block centred at 193.1 THz, using centre frequency = 193.1 THz + n × 6.25 GHz, is given below.

Table 5.2 – G.694.1 channel plan for Option R (4 × 400G, 75 GHz slots)

Channel	n	Centre frequency (THz)	Wavelength (nm)	Slot edges (THz)
1	-18	192.9875	1553.47	192.950–193.025
2	-6	193.0625	1552.87	193.025–193.100
3	6	193.1375	1552.26	193.100–193.175
4	18	193.2125	1551.66	193.175–193.250

Measurements to repeat after the upgrade: OTDR (100 ns bidirectional) and power-meter loss to show the plant is unchanged at ≤ 21.25 dB; OSA for channel frequency, power and OSNR (target well above 26 dB); pre-FEC BER and Q/constellation from the module with the post-FEC result error-free; a 24-hour BER soak test; and CD and PMD measurement.

Design Decision 4: I recommend repairing first and then upgrading to Option R (4 × 400G Si-photonic pluggables on 75 GHz slots), because it carries 1.6 Tb/s on the same fibre pair in 300 GHz (5.33 b/s/Hz), uses only 60 W (37.5 pJ/bit against 220 for P and 100 for Q), and the repaired link OSNR of 47 dB leaves 21 dB above its 26 dB requirement, while the repaired loss of 21.25 dB, PMD of 0.775 ps and absence of strong reflections keep the plant compatible with coherent detection.

Stage 6 – Verification and Critical Evaluation

Table 6.1 – Final verification table

Requirement	Specified / target	As-built	Achieved after repair / evidence
Total optical link loss	≤ 21.25 dB (design)	22.52 dB	21.25 dB after replacing event 3 (−0.70 dB) and event 2 (−0.57 dB)
Received optical power	≥ −24 dBm plus 3 dB margin	−21.52 dBm (effective −23.62 dBm)	−18.25 dBm at +3 dBm launch (effective −20.35 dBm)
System margin	≥ 3 dB	0.38 dB (Ch 3), −1.74 dB (Ch 5)	3.65 dB (all channels once Ch 5 ER is fixed)
OSNR	≥ 20 dB (10G); ≥ 26 dB (Option R)	47 dB	47 dB; margin 27 dB (10G) and 21 dB (Option R)
PMD	≤ 10 ps	0.775 ps	0.775 ps (unchanged by repair)
OTDR event condition	No event above allowance	Event 2 0.62 dB, event 3 1.20 dB, −22 dB	Event 3 ≤ 0.5 dB with ≤ −50/−60 dB reflectance; event 2 ≤ 0.05 dB, non-reflective
Eye / Q / jitter / ER	Q ≥ 7.03; ER ≥ 8.2 dB	Ch 3: Q 7.6, ER 9.0, eye 0.65 UI. Ch 5: Q 4.7, ER 4.5, eye 0.39 UI	Ch 5 target after transmitter fix: ER ≥ 9 dB, Q ≈ 7.6 or better, eye width ≈ 0.65 UI
BER and quantum-limit gap	BER ≤ 10^{-12}	Ch 3: 1.5 × 10^{-14}. Ch 5: 1.3 × 10^{-6}. Gap 20.6 dB	Ch 5 BER ≤ 10^{-12} after fix; gap 20.6 dB from the direct-detection limit
Upgrade spectrum and energy per bit	1.6 Tb/s, minimum pJ/bit	–	Option R: 300 GHz, 5.33 b/s/Hz, 60 W, 37.5 pJ/bit

The calculated values and the measurement interpretations are consistent with each other and with the original requirement. The excess loss found by the OTDR (0.70 + 0.57 = 1.27 dB) equals the difference between the as-built and design loss from the power meter (22.52 − 21.25 = 1.27 dB). The Channel 5 Q-factor drop is explained quantitatively by its ER penalty, and the OTDR, power-meter and eye data point to two separate root causes (shared fibre defects and a channel-specific transmitter). The values after repair are predictions based on the design

allowances, so they must be confirmed by the post-repair tests listed above before the upgrade is released.

Final System Design

Final parameter and diagnostic table

Item	Value / decision adopted
Fibre and length	60 km G.652.D, 0.21 dB/km, fibre loss 12.60 dB
Loss: design / as-built / repaired	21.25 dB / 22.52 dB / 21.25 dB
Excess loss found	1.27 dB (event 3: 0.70 dB, event 2: 0.57 dB)
Launch power	+3 dBm per channel (from +1 dBm)
Penalties (healthy channel)	Dispersion 1.00 dB, ER 1.10 dB; Channel 5 ER penalty 3.22 dB before repair
System margin	3.65 dB after repair and launch change
OSNR / PMD	47 dB (0.1 nm) / 0.775 ps
OTDR settings	100 ns bidirectional for the full link; 10 ns from the far end for events 3 and 4; launch and receive cables 500 m to 1 km
Event 3 and 4 repair	Priority 1: replace with LC/APC (or cleaned LC/UPC)
Event 2 repair	Priority 2: re-splice by fusion
Channel 5 repair	Re-bias or replace transmitter so ER ≥ 9 dB
BER (Ch 3 / Ch 5 as found)	1.5×10^{-14} / 1.3×10^{-6}
Eye width at 10^{-12} (Ch 3 / Ch 5)	0.651 UI / 0.390 UI
Quantum-limit gap (direct detection)	20.6 dB (homodyne 24.0 dB, heterodyne 21.0 dB)
Selected upgrade	Option R, 4 × 400G, 75 GHz slots, 300 GHz total, 5.33 b/s/Hz, 37.5 pJ/bit, 60 W
Channel plan	Centres 192.9875, 193.0625, 193.1375, 193.2125 THz

Diagnostic and upgrade workflow

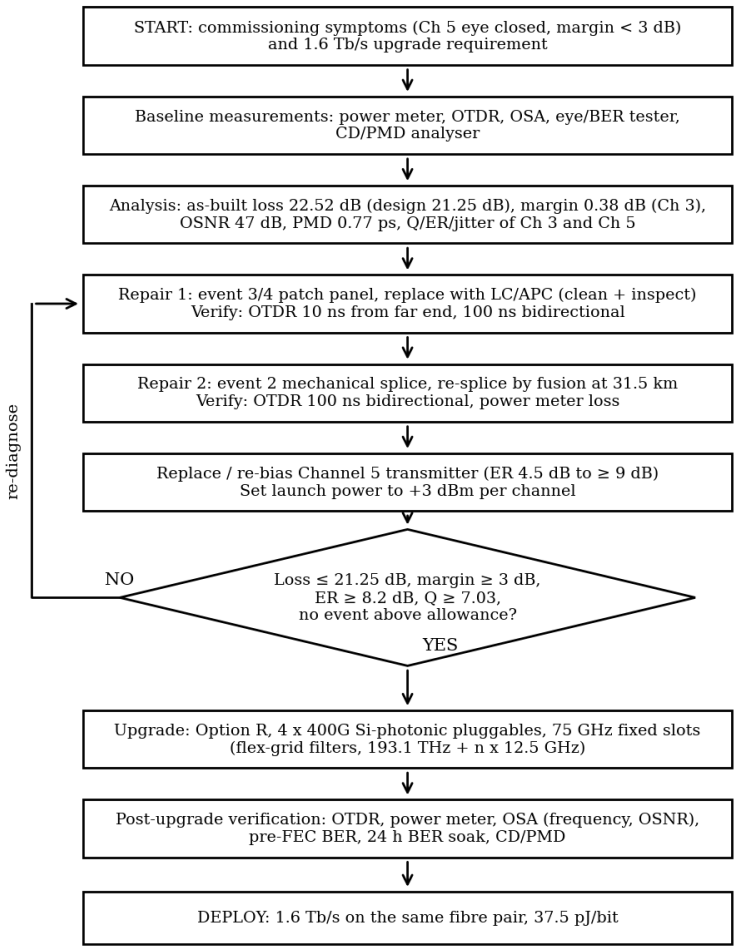


Figure 1 – Diagnostic, repair and upgrade workflow

The link budget, penalties, OSNR and margin are given in Tables 2.1 to 2.3, the OTDR events, settings and priorities in Tables 3.1 to 3.4, the eye, jitter, Q, BER and quantum-limit results in Section 4, and the G.694.1 plan, spectral efficiency and energy per bit in Tables 5.1 and 5.2.

Final engineering recommendation

Repair events 3/4 and event 2, correct the Channel 5 transmitter, raise the launch power to +3 dBm per channel so the 10G service has 3.65 dB margin, and then upgrade the same fibre pair to Option R (4 × 400G Si-photonic pluggables at 75 GHz).

Final Engineering Decision

Would I deploy the optimized design I created? YES, for the repaired and upgraded design, and only after the post-repair tests pass. I would **not** deploy the link as built.

- **Budget and margin:** the as-built margin is only 0.38 dB (Channel 3) and negative for Channel 5, so the link fails. After repair and +3 dBm launch the loss is 21.25 dB and the margin is 3.65 dB, above the 3 dB requirement, with PMD at 0.775 ps.

- **Dominant excess loss and OTDR:** event 3 (0.70 dB excess, −22 dB reflectance) and event 2 (0.57 dB excess) are the two defects. The 100 ns pulse covers the 22.52 dB range needed and the 10 ns far-end trace measures the second connector, so both repairs can be verified.
- **Eye, Q and BER:** Channel 5 (Q = 4.7, BER 1.3×10^{-6}) fails because of its transmitter ER (4.5 dB), as the Q ratio and ER penalty both show. With ER ≥ 9 dB it should behave like Channel 3 (BER 1.5×10^{-14}, below the 10^{-12} target).
- **Quantum limit and coherent detection:** the receiver is 20.6 dB from the direct-detection limit, and coherent detection is shot-noise limited, so the coherent upgrade is physically well supported with about 11.7 dB of power above the 400G homodyne limit.
- **Upgrade:** Option R gives 1.6 Tb/s in 300 GHz (5.33 b/s/Hz) at 37.5 pJ/bit, and its 26 dB OSNR requirement is met with 21 dB to spare. This is reliable for 1.6 Tb/s over 60 km without re-laying fibre.

The conditions for deployment are: end-to-end loss ≤ 21.25 dB, event 3 ≤ 0.5 dB with correct reflectance, event 2 ≤ 0.05 dB, Channel 5 with ER ≥ 8.2 dB and Q ≥ 7.03, and flex-grid filters installed for the 75 GHz slots. If any of these fails, the corrective action is to re-clean or replace the connector, redo the splice, or replace the Channel 5 transmitter, and then repeat the test. The remaining risk is that the post-repair values are predictions, which is why the verification step is a hold point.